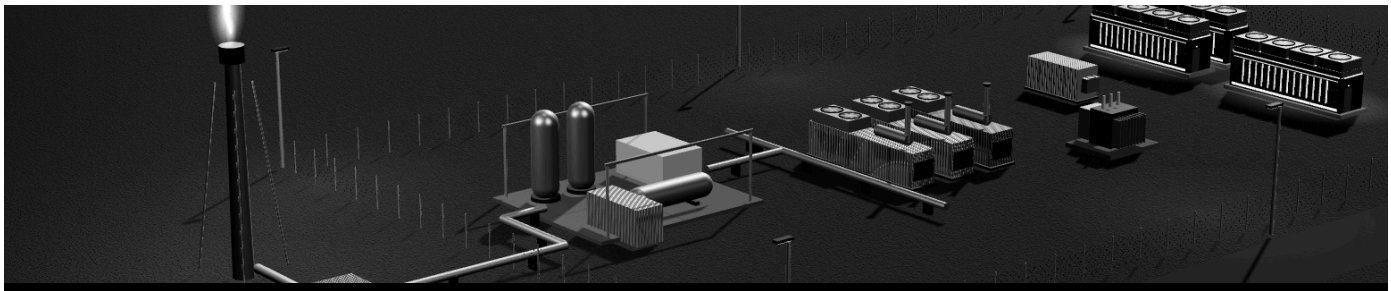
Peoria City/County Landfills, Brimfield, Illinois: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

3.1 MW	3	9 ac	283k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
$424k to $707k	$9k to $17k	$433k to $724k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=1304&CH4=43

THE MATH

Landfill gas collected and flared (2022)	1,304 scfm
Methane content (2022)	42.9%
Energy in the gas	34.0 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 3.1 MW
Compute (power less cooling and losses, /1.2)	about 2.6 MW
Compute modules (about 1 MW each)	3
Gas engines (running plus one spare)	3 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 9 acres
Gas bought per year (95% run time)	283k MMBtu
Gas payment at $1.50 to $2.50 per MMBtu	$424k to $707k
Land lease at $1,000 to $2,000 per acre per year	$9k to $17k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed or floored price.

WORTH CHECKING

- Flow now that Landfill #2 has stopped taking waste
- Status of the earlier engine project
- Who holds the gas rights
- Space for a small fenced parcel near the flare
- Gas quality (siloxanes, H2S) for engines

Source: EPA Landfill Methane Outreach Program (LMOP) database, September 2024 release (2022 data): gas collected and flared. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.